LTIMindtree

AIRLINE CASE STUDY

PROJECT GLADIATOR

SNOWFALKE GET-1066

TEAM MEMBERS

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PROBLEM DEFINITION

TO DEVELOP A SNOWFALKE APPLICATION IN ORDER TO PREDICT CANCELLATION OF COMMERCIAL FLIGHTS

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Knowing your data

1. Delimiter : Comma separated, CSV format
2. Source: Flight Cancellation Data
3. Frequent Compression data
4. Critical data

Airlines
- Columns - 2
- Rows - 31

Airports
- Columns - 7
- Rows - 322

Flights
- Columns - 14
- Rows – 10,48,497

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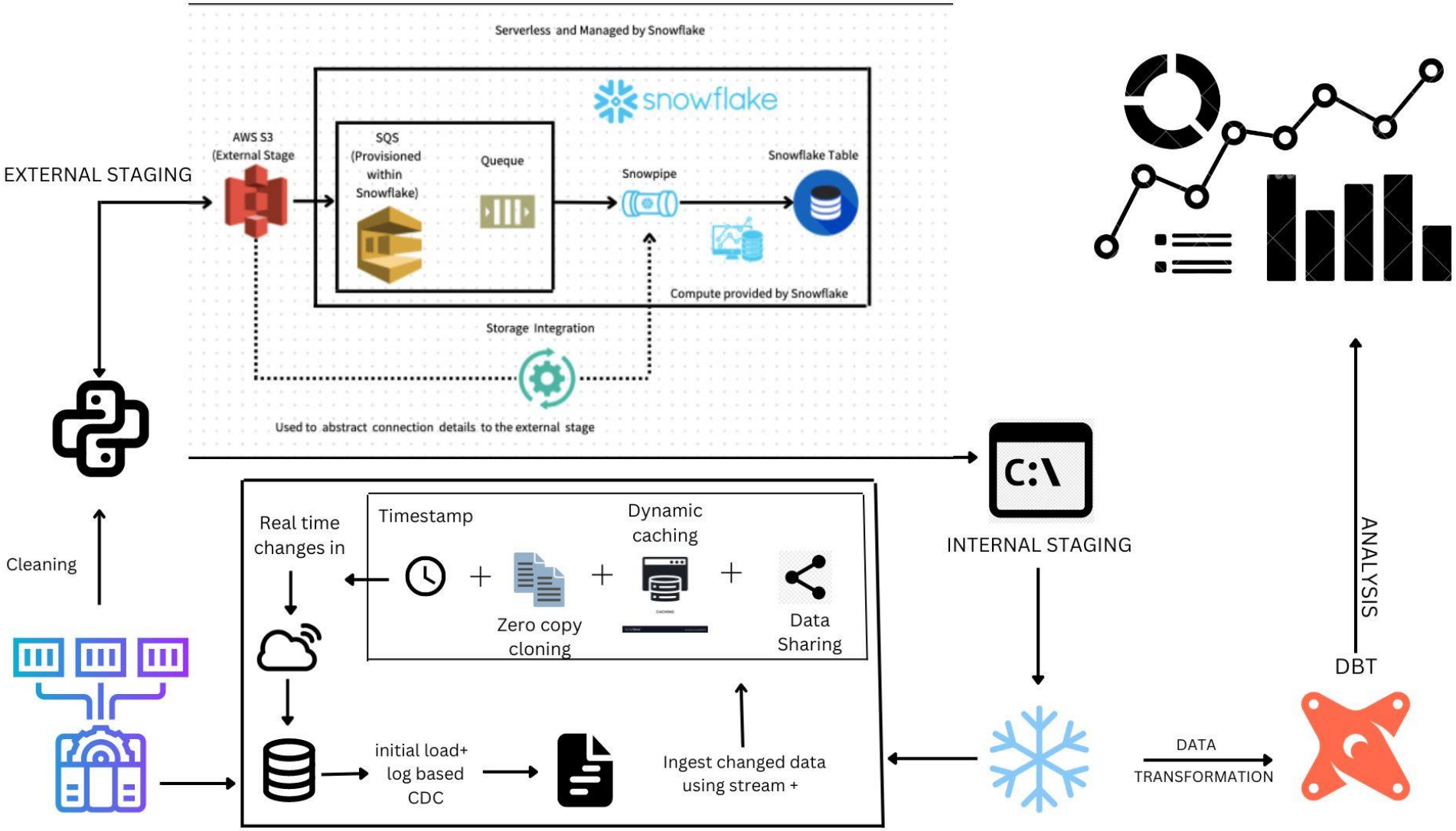
ARCHITECTURE



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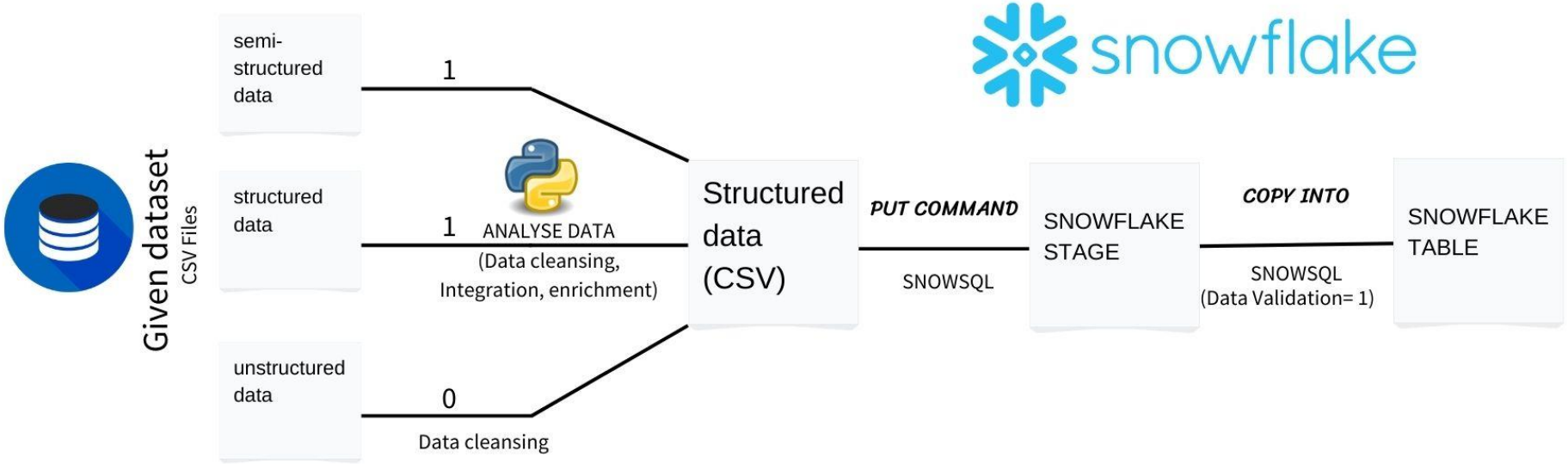
Workflow Loading from Internal Staging



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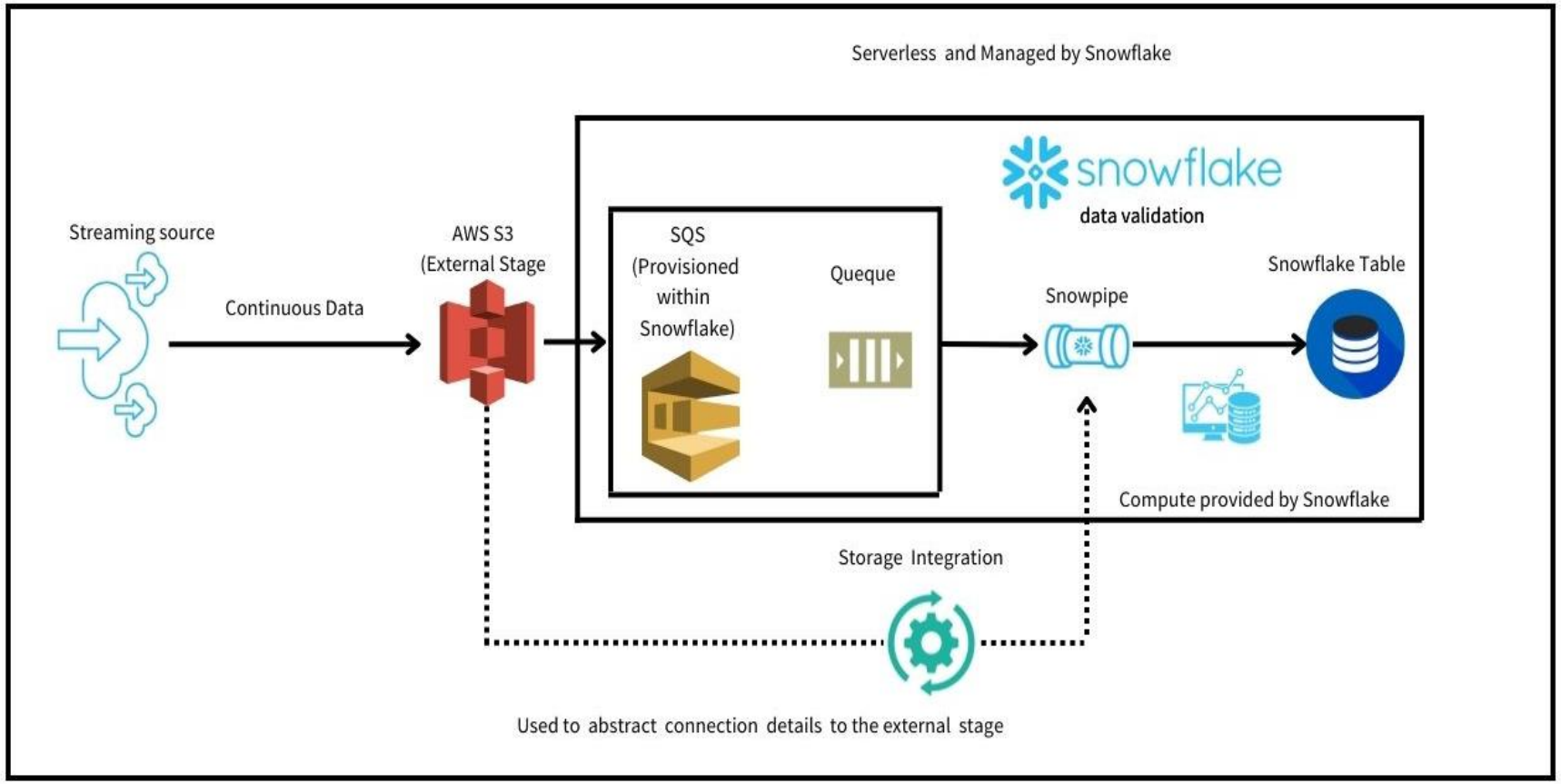
External Staging



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SNOWPIPE

1. Cloud Platform Support

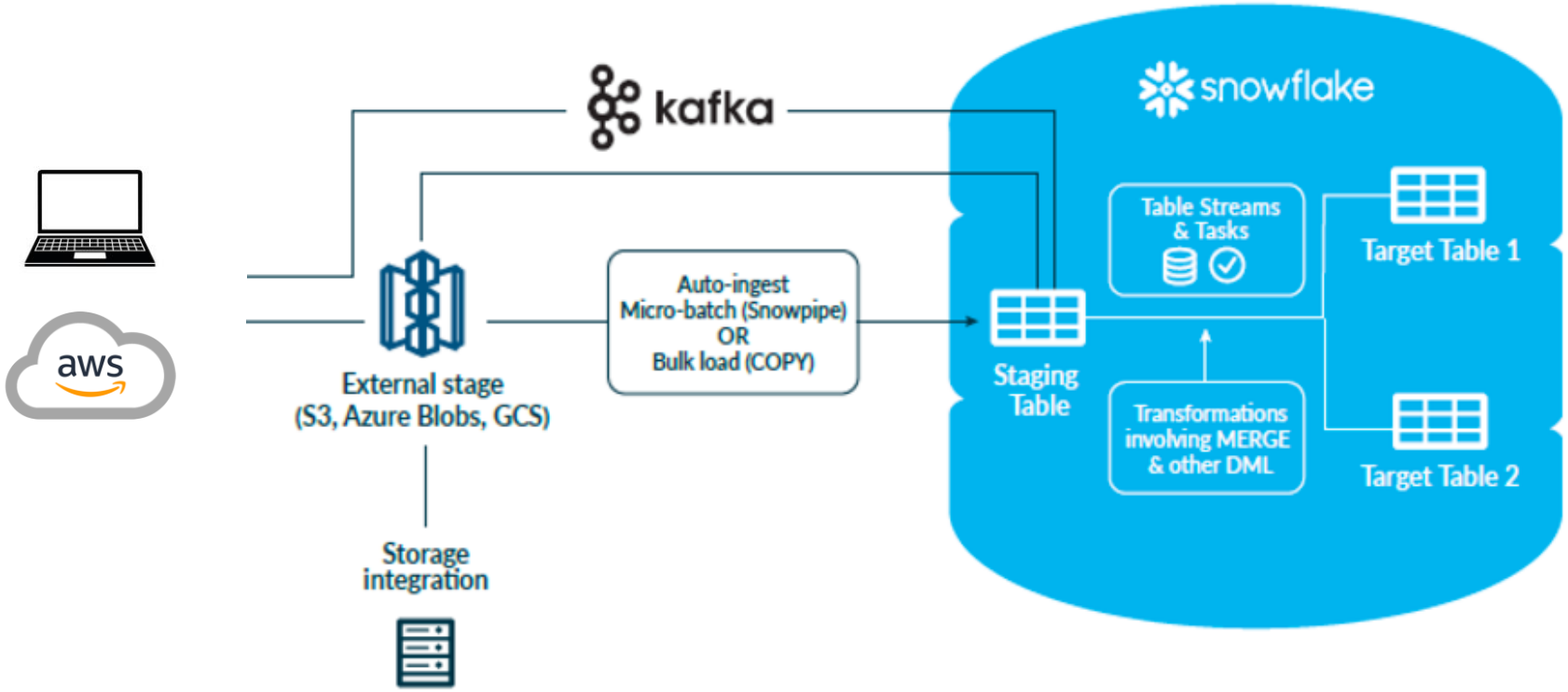
2. Network Traffic

3. Configuring Secure Storage Access

4. Creating New S3 Event Notification to Automate

OR

5. Configuring Amazon SNS with SQS



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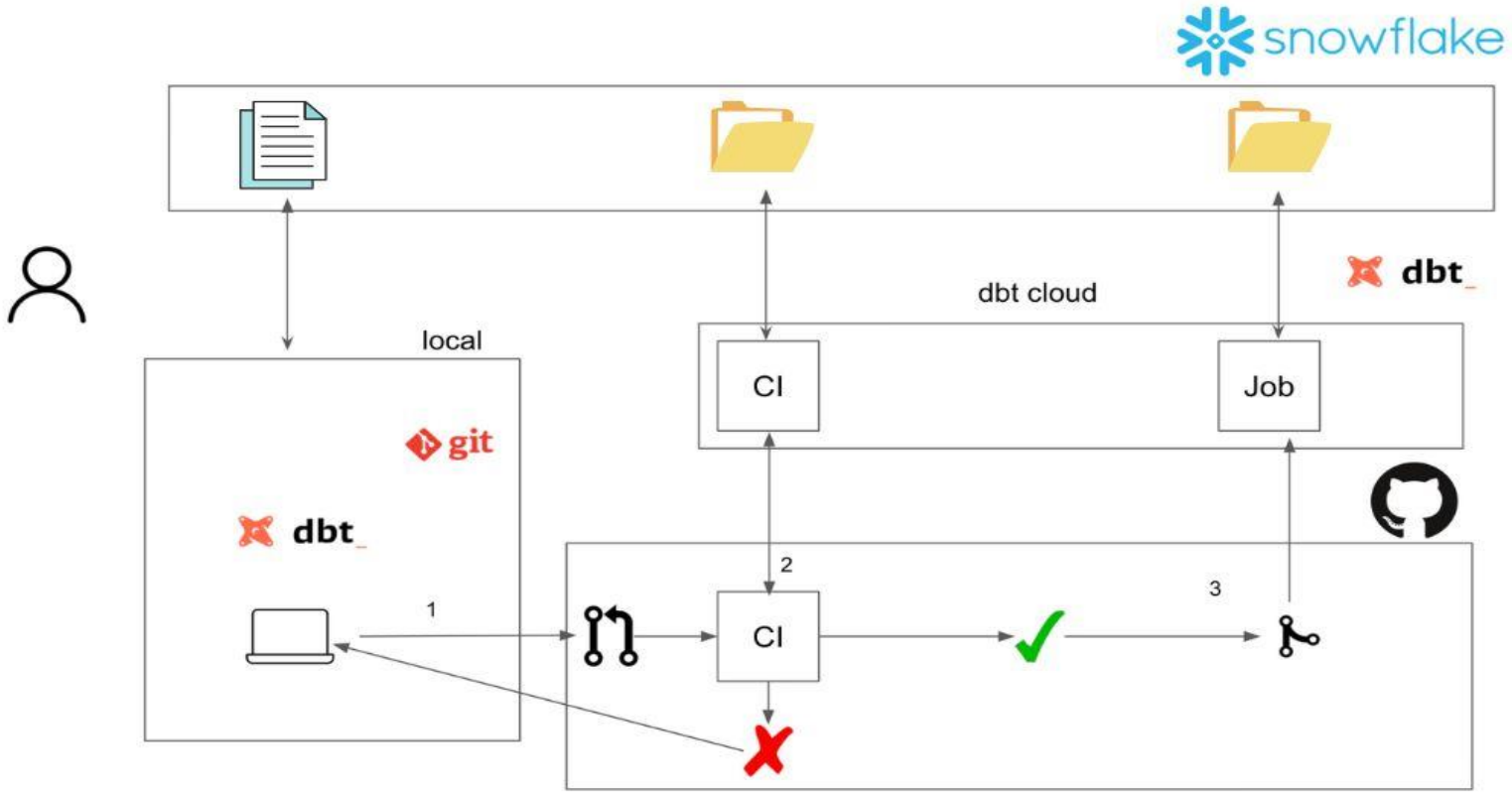
Snowflake with DBT CI/CD

1. Snowflake Configuration
2. Connect to Data Source
3. Dbt Configuration
4. Dbt Project Configuration
5. Building dbt data pipelines for Staging and Intermediate
6. Dbt pipelines- Tests & docs
7. Dbt pipelines- Deployment



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Dbt Staging

1. FLIGHTS

2. AIRLINES

3. AIRPORTS

4. sources.yml

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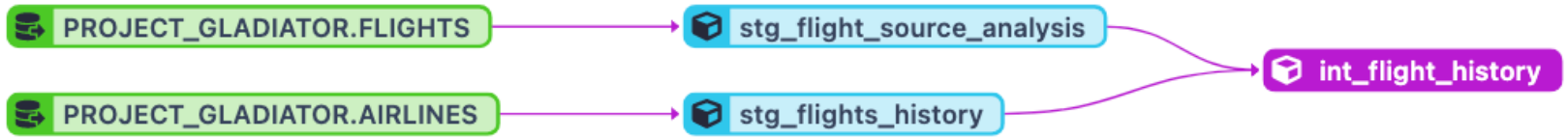
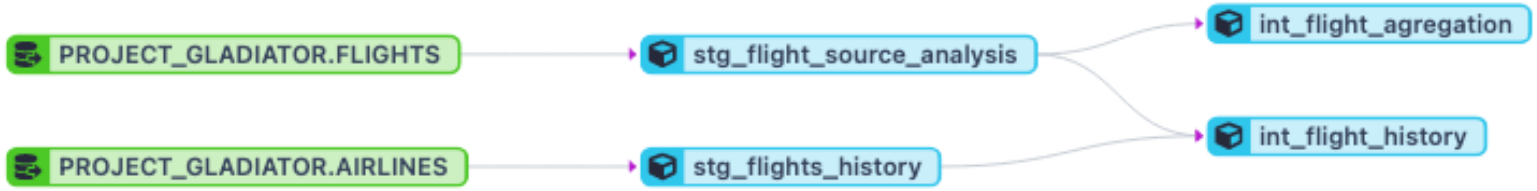
Dbt Intermediate

1. FLIGHTS



2. AIRLINES

1. intermediate.yml



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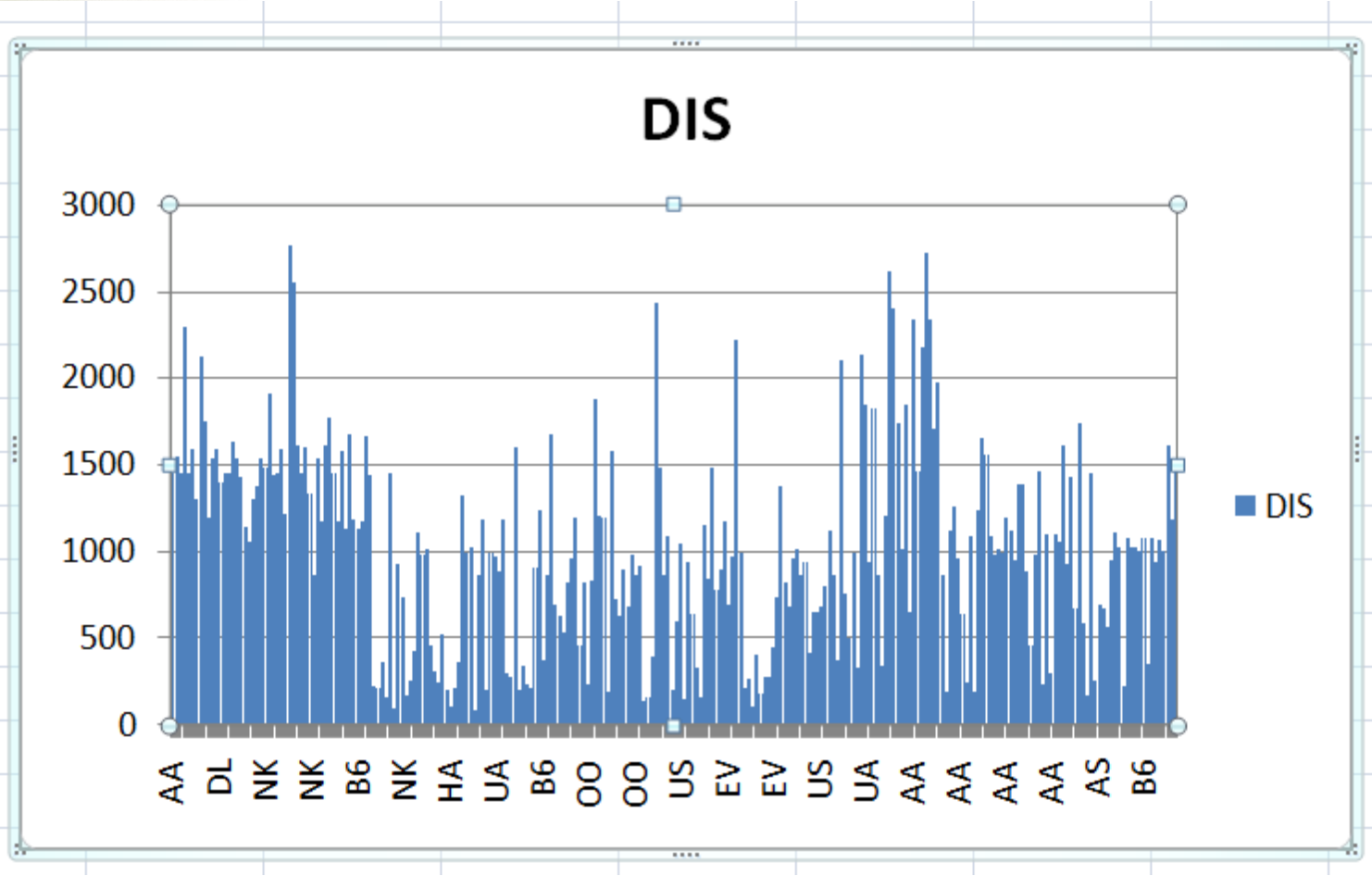
VISUALIZATION FOR DISTANCE V/S COUNTRY_CODE



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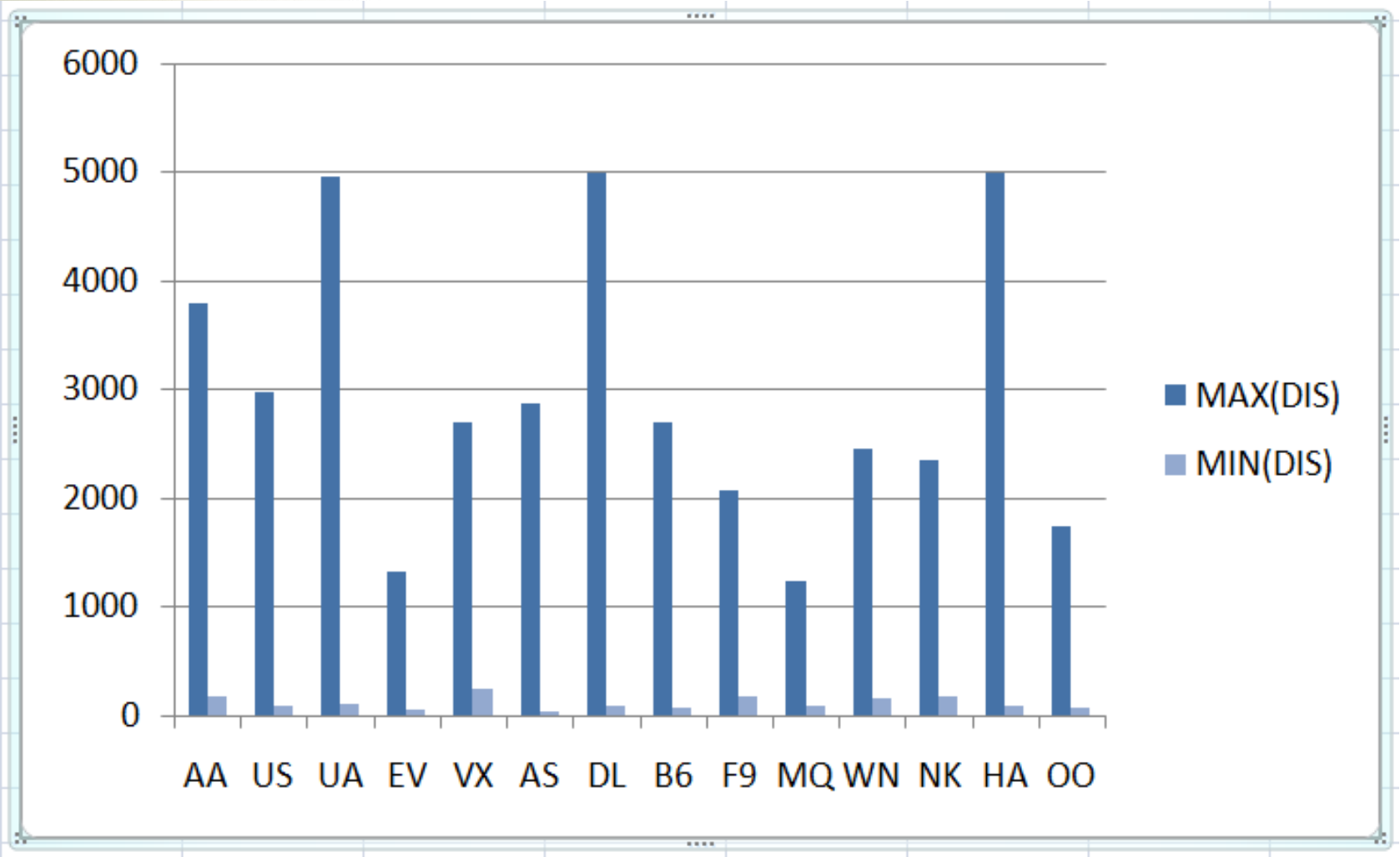
• VISUALIZATION FOR MAX,MIN DISTANCE V/S COUNTRY_CODE



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Challenges and Findings

- Issues found while creating SQS(event notification) and role.
- We have created a consumption table by merging .
- We faced issues while creating joins.
- Aggregation between origin and destination is performed to find the distance and delay.

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CONCLUSION

We learned how to use Snowflake to stream and build snowpipes. Also, connecting AWS cloud and performing Dbt to transform data using pipelines for analytics.

Using the data we can predict with the flight distance and accordingly cancellation of commercial flights are initiated.

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THANK YOU !

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